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Inventor(s)	Richards; Paul Williams

Base connections and structures including the base connections, kits for forming and methods of repairing the same

Abstract

Embodiments are directed to base connections, structures including base connections, kits for forming the base connections and/or structures, and method of repairing yielded base connections. In an embodiment, a base connection includes a base plate including a top surface and a bottom surface opposite the top surface. The top surface is configured to be adjacent to a terminal end of a column and the bottom surface adjacent to a foundation. The connection also includes one or more anchor rods attached to the base plate. The anchor rods are configured to secure the base plate to the foundation. The connection includes at least one structural fuse configured to be attached to the column and attached to the base plate. The fuse includes a plate with at least one cutout formed therein. The cutout is configured to form one or more yield regions that are configured to preferentially yield.

Inventors:	Richards; Paul Williams (Orem, UT)
Applicant:	Durafuse Frames, LLC (West Jordan, UT)
Family ID:	1000008818267
Assignee:	Durafuse Frames, LLC (West Jordan, UT)
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Primary Examiner: A; Phi D

Assistant Examiner: Tran; Phi D

Attorney, Agent or Firm: Ballard Spahr LLP

Background/Summary

BACKGROUND

(1) A column of a structure is generally attached to foundations through a base connection. The column and the base connection are designed to resist forces that may arise from wind or earthquake loading. The structure including the column and the base connection may be configured to resist severe earthquake loads by relying on ductility to prevent catastrophic failure of the structure. For example, parts of the structure may be configured to yield in a controlled manner to accommodate the large swaying associated with severe earthquake shaking. In some structures, the column is configured to yield at or near the base connection to accommodate the large swaying.

(2) It is difficult to repair a yielded column thus making the structure impractical to repair after the column has yielded. In other words, the structure that relies on the column yielding may result in a structure that is safe for severe earthquakes (i.e., the building will not collapse) but are not resilient (i.e., the structure may have to be demolished after the earthquake because the yielded column is difficult to repair).

SUMMARY

(3) Embodiments are directed to base connections, structures including base connections, kits for forming the base connections and/or structures, and method of repairing yielded base connections. In an embodiment, a base connection is disclosed. The base connection includes a base plate including a top surface and a bottom surface opposite the top surface. The top surface is configured to be adjacent to a terminal end of a column and the bottom surface adjacent to a foundation. The base connection also includes one or more anchor rods attached to the base plate. The one or more anchor rods are configured to secure the base plate to the foundation. The base connection also includes at least one structural fuse configured to be attached to the column and attached to the base plate. The at least one structural fuse includes a plate with at least one cutout formed therein. The at least one cutout is configured to form one or more yield regions extending therefrom that are configured to preferentially yield relative to other regions of the plate.

- (4) In an embodiment, a structure is disclosed. The structure includes a column, a foundation, and a base connection. The base connection includes a base plate including a top surface and a bottom surface opposite the top surface. The top surface adjacent to a terminal end of a column and the bottom surface adjacent to a foundation. The base connection also includes one or more anchor rods attached to the base plate. The one or more anchor rods securing the base plate to the foundation. The base connection also includes at least one structural fuse attached to the column and the base plate. The at least one structural fuse including a plate with at least one cutout formed therein. The at least one cutout is configured to form one or more yield regions extending therefrom that are configured to preferentially yield relative to other regions of the plate.
- (5) In an embodiment, a kit is disclosed. The kit includes a base plate including a top surface and a bottom surface opposite the top surface, the top surface configured to be adjacent to a terminal end of a column and the bottom surface configured to be adjacent to a foundation. The kit further includes at least one structural fuse configured to be attached to the column and the base plate. The at least one structural fuse includes a plate with at least one cutout formed therein. The at least one cutout is configured to form one or more yield regions extend therefrom that are configured to preferentially yield relative to other regions of the plate.
- (6) In an embodiment, a method of repairing a yielded structural fuse is disclosed. The method includes detaching the yielded structural fuse from a column and a base plate. The base connection include the base plate including a top surface and a bottom surface opposite the top surface. The top surface adjacent to a terminal end of the column and the bottom surface adjacent to a foundation. The base connection also includes one or more anchor rods attached to the base plate. The one or more anchor rods secure the base plate to the foundation. The base connection further includes at least one structural fuse including a plate with at least one cutout formed therein. The at least one cutout configured to form one or more yield regions extend therefrom that are configured to preferentially yield relative to other regions of the plate. The at least one structural fuse includes the yielded structural fuse. The method also includes attaching a new structural fuse to the column and the base plate. The new structural fuse includes a new plate with at least one new cutout formed therein. The at least one new cutout is configured to form one or more new yield regions extending therefrom that are configured to preferentially yield relative to other regions of the plate.
- (7) Features from any of the disclosed embodiments may be used in combination with one another, without limitation. In addition, other features and advantages of the present disclosure will become apparent to those of ordinary skill in the art through consideration of the following detailed description and the accompanying drawings.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The drawings illustrate several embodiments of the present disclosure, wherein identical reference numerals refer to identical or similar elements or features in different views or embodiments shown in the drawings.
- (2) FIGS. 1A to 1C are lateral side, top, and frontal side views of a portion of a structure, according to an embodiment.
- (3) FIG. 1D is an exploded view of the structure illustrated in FIGS. 1A to 1C.
- (4) FIG. 2 is a front elevational view of a structural fuse illustrated in FIGS. 1A to 1D.
- (5) FIG. 3A is a side elevational view of the structure illustrated in FIGS. 1A-1D, according to an embodiment.
- (6) FIG. 3B is a front elevational view of a yielded structural fuse, according to an embodiment.
- (7) FIG. 4 is a flow chart of a method for repairing the structure, according to an embodiment.
- (8) FIG. 5 is a side elevation view of a structure with a load applied thereto, according to an

embodiment.

DETAILED DESCRIPTION

(9) Embodiments are directed to base connections, structures including base connections, kits for forming the base connections and/or structures, and methods of repairing yielded base connections. An example base connection includes a base plate including a top surface and a bottom surface opposite the top surface. During use, the top surface of the base plate may be adjacent to a terminal end of a column and the bottom surface may be positioned adjacent to a foundation. The base plate further includes one or more anchor rods that are attached to the base plate and secure the base plate to the foundation. The base connection also includes at least one structural fuse that, during use, connects the column to the base plate. The structural fuse includes a plate with at least one cutout formed therein. The cutout is configured to form one or more yield regions extending therefrom.

(10) The structural fuses disclosed herein are configured to preferentially absorb and dissipate energy from a load by preferentially yielding. As used herein, “yield,” “yielding,” and “yielded” may refer to failing, fracturing, plastically deforming, damaging or otherwise yielding an element (e.g., structural fuse) in a manner that may or may not require the replacement of the element after failure. Examples of loads that may cause yielding of the structural fuses include loads caused by a seismic event or wind. The structural fuses may absorb and dissipate some of the energy of loads applied to the structure that includes the base connection which may prevent or avoid yielding of the column that may otherwise result from the load. As such, the structural fuses disclosed herein may move yielding from the column to the structural fuses.

(11) The base connection that includes the structural fuse is an improvement over a base connection that does not include the structural fuse (i.e., a base connection that includes a base plate and anchor rods). Several examples of base connections that do not include structural fuses are provided. In an example, the base connection that does not include the structural fuse may include a column welded to the base plate. In such an example, the base plate, the anchor rods, and the foundation are configured to be stronger than the column. That way, the column will yield rather than yielding the base plate, the anchor rods, or the foundation. When the column is relatively strong, a very thick base plate, numerous heavy anchor rods, and a substantial foundation are required in order to ensure that the base plate, anchor rods, and foundation are stronger than the column which may significantly increase project budgets. Regardless, the yielded column is difficult to remove and replace thereby making repairing the structure difficult or impractical after the column yields. In an example, the base connection that does not include the structural fuse may include a column welded to the base plate. In such an example, the base plate and/or the anchor rods of the base connection may be configured to yield instead of the column. However, similar to yielding the column, the base plate and the anchor rods may be difficult or impractical to repair after yielding since such repairs may require lifting the column off the base plate or removing portions of the foundation. In an example, the base connection that does not include the structural fuse may include one or more angles that attach the column to the base plate. The angles may be configured to yield. The yielded angle may be more easily repaired than if the column, the base plate, or the anchor rods yielded. However, attaching the column to the base plate using the angles results in a structure that is not stiff and is only partially restrained. In an example, the base connection that does not include the structural fuse may include at least one flange plate with slotted holes that attach the column to the base plate. In such an example, the flange plate may be bolted to the column and the bolts may slip in the slotted holes to accommodate the column swaying. In other words, the base connection relies on friction to dissipate energy. The flange plate with the slotted holes may prevent yielding of the base connection and the column thereby preventing the need to repair the base plate and the column. However, the flange plate with the slotted holes may have unreliable post-slip stiffness, poor strength, unpredictable bolt slip resistance, and the force required to cause the bolts to slip may change over time.

(12) As previously discussed, the base connections that include a structural fuse are an improvement over base connections that do not include the structural fuse. For example, the structural fuses disclosed herein are configured to yield when a sufficiently large load is applied to the structure that includes the base connection. Yielding the structural fuse may prevent yielding of the base plate, the anchor rods, and the column. Unlike yielding the column, the base plate, and the anchor rods, repairing the yielded structural fuse may be relatively simple thereby causing such as structure that includes the structural fuse to be resilient (i.e., the structure does not need to be demolished after the structural fuse yields). Configuring the structural fuse to preferentially yield prevents the need to use very thick base plates, numerous heavy anchor rods, and substantial foundations, even when the base connection is attached to a relatively strong column. Further, the structural fuse may allow the stiffness of the connection between base connection and the column to qualify as a fully-restrained connection. The fully-restrained connection allows lighter beams and columns to be used in the structure than if the connection between the base connection and the column was a partially-restrained connection while maintaining the required overall stiffness. Finally, the expected load that causes the structural fuses to yield may be easier to predict than the expected load that causes the bolts to slip in the slotted holes of the flange plate. In particular, the load that causes the structural fuse to yield may be reliably predicted and will not change over time. Thus, the base connections including at least one structural fuse is an improvement over base connections that do not include a structural fuse.

(13) FIGS. 1A to 1C are lateral side, top, and frontal side views of a portion of a structure **100**, according to an embodiment. FIG. 1D is an exploded view of the structure **100** illustrated in FIGS. 1A to 1C. The structure **100** includes a column **102**, a base connection **104**, and a foundation **106**. The base connection **104** is configured to attach the column **102** to the foundation **106**. The base connection **104** includes a base plate **108** and one or more anchoring rods **110**. The anchoring rods **110** are configured to extend into the foundation **106** thereby securing the base plate **108** to the foundation **106**. The base connection **104** also includes at least one structural fuse **112**. The structural fuse **112** is configured to be attached to the column **102** and the base plate **108** thereby attaching the column **102** and the base plate **108** together. The structural fuse **112** is configured to preferentially yield when a sufficient load is applied to the structure **100** thereby preventing or at least inhibiting yielding of the column **102**, the foundation **106**, the base plate **108**, and the anchor rods **110**. In an embodiment, when the structural fuse **112** is indirectly attached to the base plate **108**, the base connection **104** may include at least one flange plate **114** that is configured to connect the structural fuse **112** to the base plate **108**.

(14) In an embodiment, the column **102** includes an I-beam. In such an embodiment, the column **102** includes two flanges **116** with a web **118** extending between the two flanges **116**. The structural fuse **112** may be attached (e.g., bolted, riveted, welded, etc.) to one or both of the two flanges **116**. In an embodiment, the column **102** may include a structural beam other than an I-beam, such as a T-beam, an angle, a hollowed sectioned structural beam, or any other suitable structural beam.

(15) The base plate **108** includes a top surface **124** and a bottom surface **126** opposite the top surface **124**. The top surface **124** of the base plate **108** is configured to be adjacent to (e.g., directly contact) a terminal end **128** of the column **102**. Generally, the top surface **124** extends outwardly from all of the outer edges of the terminal end **128** of the column **102**. In other words, the base plate **108** is wider than the column **102**. Extending the top surface **124** outwardly from all of the outer edges of the terminal end **128** of the column **102** better distributes the weight of the column **102** and the rest of the structure **100** attached to the column **102** (not shown) across the base plate **108**. Further, extending the top surface **124** outwardly from all of the outer edges of the terminal end **128** of the column **102** provides portions of the base plate **108** in which anchor holes **130** may be formed. The bottom surface **126** is configured to be adjacent to the outer surface **132** of the foundation **106**.

(16) The base plate **108** may exhibit any suitable shape. In an example, the base plate **108** may

exhibit a generally square, a generally rectangular, a generally circular, or a generally oval shape. In some examples, the base plate **108** may not include at least one cutout formed therein that weakens a portion of the base plate **108**, similar to the cutout **140** of the structural fuse **112**, since such a cutout formed in the base plate **108** may increase the likelihood that the base plate **108** yields. As previously discussed, repairing a yielded base plate **108** may be difficult or impractical.

(17) As previously discussed, the anchoring rods **110** are configured to secure the base plate **108** to the foundation **106**. In an embodiment, as illustrated, the anchoring rods **110** are configured to extend through the anchor holes **130** formed in the base plate **108** and disposed in the foundation **106** (e.g., the foundation **106** is formed around the anchor rods **110**). The anchor rods **110** may include one or more threads formed thereon. The threads allow one or more nuts **134** to be secured to the anchor rods **110**. In an example, at least one of the nuts **134** may be disposed outside of the foundation **106**. In such an example, the nut **134** may exhibit a lateral dimension (e.g., diameter) that is greater than the lateral dimension of the anchor holes **130**. As such, the nuts **134** may be disposed on the anchor rods **110** after the anchor rods **110** are disposed through the anchor holes **130** thereby securing the base plate **108** to the anchor rods **110**. In an example, at least one of the nuts **134** are disposed in the foundation **106** which may better inhibit pullout of the anchor rods **110** from the foundation **106** than if the anchor rods **110** did not include the nuts **134** disposed in the foundation **106**.

(18) The base connection **104** may include any suitable number of anchor rods **110**. For example, in the illustrated embodiment, the base connection **104** may include 6 anchor rods **110**. However, the base connection **104** may include more than 6 anchor rods **110** (e.g., 7, 8, 9, 10, 11, 12, or more than 12) or fewer than 6 anchor rods **110** (e.g., 1, 2, 3, 4, or 5). As previously discussed, configuring the structural fuse **112** to preferentially yield allows the base connection **104** to include fewer anchor rods **110** than if a similar base connection was used that did not include the structural fuse **112**, all other conditions the same, since the presence of the structural fuse **112** does not require the anchor rods **110** to be configured to resist yielding (e.g., the anchor rods **110**, collectively, do not need to be stronger than the column **102**, the base plate **108**, etc.). The fewer anchor rods **110** makes forming the base connection **104** easier and cheaper.

(19) The structural fuse **112** is configured to be attached to the column **102** and the base plate **108**. As such, as best shown in FIG. 2, which is a front elevational view of the structural fuse **112** illustrated in FIGS. 1A-1D, the structural fuse **112** may include one or more attachment regions that are configured to be attached to the column **102** and/or the base plate **108**. In an embodiment, as illustrated, the structural fuse **112** includes at least one first connection region **136** and at least one second connection region **138**. For illustrative purposes, the first and second connection regions **136**, **138** are illustrated in FIG. 2 using diagonal intersecting lines. The first connection region **136** is configured to be attached to the column **102** and the second connection region **138** is configured to be attached (e.g., directly or indirectly attached), to the base plate **108**. The first and second connection regions **136**, **138** may be attached to the column **102** and the base plate **108**, respectively, using any suitable technique. In an example, as illustrated, the first connection region **136** is configured to be attached to the column **102** using one or more bolts **120**. As such, both the first connection region **136** and a portion of the column **102** configured to be attached to the structural fuse **112** includes one or more bolt holes **122** through which the bolts **120** may be disposed. Configuring the structural fuse **112** to be attached to the column **102** using the bolts **120** makes removal of the structural fuse **112** easier when the structural fuse **112** yields and needs to be replaced. In an example, the first connection region **136** may be attached to the column **102** via welding. Welding the first connection region **136** to the column **102** may make the connection between the base connection **104** and the column **102** more stiff but may make removing a yielded structural fuse (shown in FIG. 3B) more difficult. In an example, the second connection portion **138** may be attached to the base plate **108** indirectly using at least one flange plate **114**, as will be discussed in more detail below. The second connection portion **138** may be configured to be

attached to the flange plate **114** using one or more bolts **120** or rivets which may facilitate repair of the structural fuse **112** or one or more welds which may make the connection between the base connection **104** and the column **102** more stiff but may make repairing the structural fuse **112** more difficult. In an example, the second connection portion **138** may be directly welded to the base plate **108** which may make the connection between the base connection **104** and the column **102** more stiff but may make repairing the structural fuse **112** more difficult. It is noted that the structural fuse **112** may be attached to the column **102** and the base plate **108** (e.g., to the flange plate **114**) using rivets or other attachment techniques instead of or in addition to bolts **120** or welds.

(20) In an embodiment, the structural fuse **112** directly contacts the column **102**. Directly contacting the structural fuse **112** to the column **102** may allow a spacer to be omitted from the base connection **104** that would otherwise need to be positioned between the structural fuse **112** and the column **102**. Omitting the spacer decreases the complexity and cost of attaching the base connection **104** to the column **102**. Omitting the spacer may also increase the stiffness of the connection between the column **102** and the base connection **104** (e.g., between the column **102** and the structural fuse **112**). In other words, omitting the space may make the connection between the column **102** and the base connection **104** more fully restrained. In an embodiment, the structure **100** includes at least one spacer (not shown) between the structural fuse **112** and the column **102**.

(21) The structural fuse **112** includes at least one plate **139**. The structural fuse **112** includes at least one cutout **140** formed in the plate **139**. The cutout **140** is configured to weaken the plate **139** such that the plate **139** yields in selected regions of the plate **139**. For example, the cutout **140** is configured such that the plate **139** yields in one or more yield regions **142** (illustrated in FIGS. **1D** and **2** using non-intersecting lines). At least a portion (e.g., majority or all) of the yield regions **142** are distinct from at least a portion (e.g., majority or all) of the connection regions. As such, failure of plastic deformation of the yield regions **142** of the plate **139** are unlikely to affect the connection between the plate **139** and the other elements of the structure to which the structural fuse **112** is attached. In some embodiments, at least one of the cutout **140** and the yield regions **142** are positioned between the first and second connection regions **136**, **138** which may facilitate yielding of the structural fuse **112**.

(22) In an embodiment, the cutout **140** may include an opening formed in the plate **139** that extends through the plate **139**. The cutout **140** does not attach the structural fuse **112** to the column **102** or the flange plate **114**. When the cutout **140** extends through the plate **139**, the cutout **140** is distinguishable from the bolt holes **122** that are configured to receive the bolts **120** or rivets by the size of the cutout **140**. In an example, the cutout **140** exhibits a maximum length or area that is significantly larger (e.g., at least 2 times larger, at least 5 times larger, or at least 10 times larger) than the bolt holes **122** and the maximum length may be about 1.5 cm or greater (e.g., about 2 cm or greater, about 3 cm or greater, about 4 cm or greater, about 5 cm or greater, about 7.5 cm or greater, or 10 cm or greater). The cutout **140** may be significantly larger than the bolt holes **122** since the cutout **140** is configured to selectively weaken the plate **139** whereas the bolt holes **122** are configured to have a negligible effect on the strength of the plate **139**. In an example, the cutout **140** may be distinguishable from the bolt holes **122** because the cutout **140** exhibits a non-circular shape (e.g., elongated or square shape) while the bolt holes **122** are circular. In an embodiment, the cutout **140** may include a selectively thinned region of the plate **139**.

(23) The cutout **140** may extend inwardly from an edge **144** of the plate **139** or may be completely surrounded from the plate **139**. In an example, the at least one cutout **140** may include a single cutout **140** that is completely surrounded by the plate **139**. In an example, the at least one cutout **140** includes a single cutout **140** that extends inwardly from one edge **144** of the plate **139**. In an example, the at least one cutout **140** includes a plurality of cutouts **140**. In such an example, the plurality of cutouts **140** may extend inwardly from at least one edge **144** of the plate **139**, be completely surrounded by the plate **139**, or both (e.g., at least one cutout **140** extends inwardly

from the edge **144** while at least one other cutout **140** is surrounded by the plate **139**).

(24) The cutout **140** may exhibit any suitable shape. Generally, the shape of the cutout **140** exhibits a generally rounded shape, such as circular or oblong shape, to prevent stress concentrators which may cause the structural fuse **112** to fail or plastically deform at unsatisfactory low loads. However, the cutout **140** may exhibit a non-rounded shape, such as a rectangular or square shape. The stress concentrators (e.g., corners) of such non-rounded shapes may allow for more control over which portions of the plate **139** are the yield regions **142**. In an example, the cutout **140** may exhibit a longitudinally extending shape, such as an oblong, ellipsoid, or rectangular shape. The longitudinally extending shape may weaken region of the plate **139** that are aligned with the longitudinal axis of the longitudinally extending shape of the cutout **140** thereby allowing for more control of which portions of the plate **139** are the yield region **142**. That is, the yield regions **142** are the portions of the plate **139** that are aligned with the longitudinal axis of the cutout **140**.

(25) As previously discussed, the at least one cutout **140** may include a plurality of cutouts **140**. In an embodiment, at least some of the plurality of cutouts **140** may be arranged on the plate **139** in at least one generally straight line. Arranging the plurality of cutouts **140** in a generally straight line causes the yield region **142** to be aligned and positioned on the generally straight line. As such, arranging the plurality of cutouts **140** in a generally straight line may allow for better control of which portions of the plate **139** that are yield region **142**. However, as illustrated in FIGS. 3A and 3B, at least one of the plurality of cutouts **140** do not have to be arranged in a generally straight line.

(26) As previously discussed, the plate **139** includes at least one yield region **142**. The yield region **142** are portions of the plate **139** that are weakened by the cutout **140** such that the yield region **142** preferentially yield when a load is applied to the plate **139**. In an example, the at least one cutout **140** includes a plurality of cutouts **140** and the yield region **142** is between adjacent ones of the cutouts **140**. In such an example, the yield region **142** is between the adjacent cutouts **140** because the adjacent cutouts **140** weaken a portion of the plate **139** between the cutouts **140**. For instance, the yield region **142** is between the cutout **140** and the edge **144** of the plate **139** nearest the cutout **140**.

(27) The direction that the yield region **142** extends from the cutout **140** effects which load applied to the structural fuse causes the yield region **142** to yield. For example, only loads that are generally parallel to the direction that the yield region **142** extends from the cutout **140** may cause the yield region **142** to yield. When a load is applied to the structural fuse **112** that is obliquely angled relative to the yield region **142**, the obliquely angled load may be broken into a first load component that is generally parallel to the direction that the yield region **142** extends from the cutout **140** and a second load component that is perpendicular to the first load. The first load component may cause the yield region **142** to yield while the second load component is unlikely to cause the yield region **142** to yield.

(28) The strength of the structural fuse **112** (e.g., the load that the structural fuse **112** may withstand without yielding) may be adjusted by controlling the thickness of the plate **139**, the spacing d between adjacent cutouts **140**, the length L of the cutouts **140** that is the maximum lateral dimension of the cutouts **140**, and the width W of the cutouts **140** measure perpendicularly to the length L . Generally, increasing the thickness, increasing the spacing d , decreasing the length L , and decreasing the width W increase the strength of the structural fuse **112**, and vice versa. In an example, the thickness of the plate **139** may be selected to be about 0.25 cm or greater, about 0.5 cm or greater, about 0.75 cm or greater, about 1 cm or greater, about 1.25 cm or greater, about 1.5 cm or greater, about 2 cm or greater, about 2.5 cm or greater, about 3 cm or greater, about 4 cm or greater, about 5 cm or greater, about 6 cm or greater, about 7 cm or greater, about 8 cm or greater, or in ranges of about 0.25 cm to about 0.75 cm, about 0.5 cm to about 1 cm, about 0.75 cm to about 1.25 cm, about 1 cm to about 1.5 cm, about 1.25 cm to about 2 cm, about 1.5 cm to about 2.5 cm, about 2 cm to about 3 cm, about 2.5 cm to about 4 cm, about 3 cm to about 5 cm, about 4 cm to

about 6 cm, about 5 cm to about 7 cm, or about 6 cm to about 8 cm. In an example, the spacing **d** may be selected to about 2 cm or greater, about 3 cm or greater, about 4 cm or greater, about 5 cm or greater, about 6 cm or greater, about 7 cm or greater, about 8 cm or greater, about 10 cm or greater, about 12.5 cm or greater, about 15 cm or greater, about 20 cm or greater, about 25 cm or greater, or in ranges of about 2 cm to about 4 cm, about 3 cm to about 5 cm, about 4 cm to about 6 cm, about 5 cm to about 7 cm, about 6 cm to about 8 cm, about 7 cm to about 10 cm, about 8 cm to about 12.5 cm, about 10 cm to about 15 cm, about 12.5 cm to about 20 cm, or about 15 cm to about 25 cm. In an example, the width **W** of the cutouts **140** may be selected to be about 1 cm or greater, about 1.5 cm or greater, about 2 cm or greater, about 2.5 cm or greater, about 3 cm or greater, about 4 cm or greater, about 5 cm or greater, about 6 cm or greater, about 7.5 cm or greater, about 10 cm or greater, about 12.5 cm or greater, about 15 cm or greater, about 20 cm or greater, about 25 cm or greater, or in ranges of about 1 cm to about 2 cm, about 1.5 cm to about 2.5 cm, about 2 cm to about 3 cm, about 2.5 cm to about 4 cm, about 3 cm to about 5 cm, about 4 cm to about 6 cm, about 5 cm to about 7.5 cm, about 6 cm to about 10 cm, about 7.5 cm to about 12.5 cm, about 10 cm to about 15 cm, about 12.5 m to about 20 cm, or about 15 cm to about 25 cm.

(29) Referring back to FIGS. **1A-1D**, as previously discussed, the base connection **104** may include at least one flange plate **114** that attaches the structural fuse **112** to the base plate **108** though, it is noted, the flange plate **114** may be omitted. When the flange plate **114** is omitted, the structural fuse **112** may be attached to the base plate **108** via welding, using bolts or rivets (e.g., the structural fuse **112** is an angle), or any other suitable technique.

(30) The flange plate **114** is configured to be attached to or integrally formed with the base plate **108**. The flange plate **114** is also configured to be attached to the structural fuse **112**. The flange plate **114** may be attached to the base plate **108** and the structural fuse **112** using any suitable technique. In an example, as illustrated, the flange plate **114** is configured to be attached to the base plate **108** via welding which may make the base connection **104** more stiff. However, it is noted that the flange plate **114** may be attached to the base plate **108** using bolts or rivets (e.g., the flange plate **114** is an angle) or any other suitable techniques. In an example, as illustrated, the flange plate **114** may be attached to the structural fuse **112** using one or more bolts **120** or rivets. In such an example, the flange plate **114** may define one or more bolt holes **122** that correspond to one or more of the bolt holes **122** formed in the structural fuse **112** (e.g., the bolts holes **122** of the second connection portion **138**). Connecting the flange plate **114** to the structural fuse **112** with bolts **120** or rivets may facilitate removal of the structural fuse **112** from the base connection **104** when the structural fuse **112** yields. However, it is noted that that the flange plate **114** may be welded or otherwise attached to the structural fuse **112**. In an embodiment, the flange plate **114** may be omitted from the base connection **104** when, for example, the structural fuse **112** is directly attached to the base plate **108**.

(31) The flange plate **114** may define an opening **146**. The opening **146** may extend from an edge **148** of the flange plate **114**. The flange plate **114** may exhibit a generally U-like shape when the opening **146** extends inwardly from the edge **148**. Alternatively, the flange plate **114** may completely enclose the opening **146**. The opening **146** of the flange plate **114** may be configured such that the first connection region **136** of the structural fuse **112** is not covered by the flange plate **114**. As such, the opening **146** may allow access to the first connection region **136** when the structural fuse **112** is attached to the flange plate **114** and allow the structural fuse **112** to be attached or detached from the column **102**. The opening **146** may also allow the structural fuse **112** to be attached to the column **102** with bolts **120** or other attachment mechanisms that would protrude outwardly from the structural fuse **112** without needing to form a recess in the flange plate **114** to accommodate such attachment mechanisms. In an embodiment, the flange plate **114** may define at least one recess that is configured to accommodate bolts **120**, rivets, or another attachment mechanism that extend outwardly from the structural fuse **112** instead of the opening **146**.

(32) In an embodiment, the connection between the column **102** and the base connection **104** is

sufficiently stiff to qualify as fully restrained connection. The fully restrained connection may allow for smaller columns to be attached to the base connection **104** than if the connection between the column **102** and the base connection **104** was only a partially restrained connection. In an embodiment, the connection between the column **102** and the base connection **104** is only a partially restrained connection.

(33) In an embodiment, as previously discussed, a spacer (e.g., plate) between the column **102** and the structural fuse **112** may be omitted from the structure **100**. Such a spacer may be omitted since the structural fuses disclosed herein are unlikely to buckle when in compression. Further, the spacer may be omitted since the structural fuse **112** directly contact the column **102** which makes the connection between the column **102** and the structural fuse **112** more stiff. In an embodiment, a cover plate that covers all or substantially all of the structural fuse **112** and is distinct from the flange plate **114** may be omitted from the structure **100**. The cover plate may be omitted because the structural fuses disclosed herein are unlikely to buckle when compressed and the cover plate may interfere with the flange plate **114**. In an embodiment, one or more post-tensioned bars may be omitted from the structure **100** since such post-tensioned bars are not necessary. In an embodiment, flange plates with slotted holes may be omitted from the structure **100** since the structural fuse **112** is configured to yield thereby absorbing and dissipating energy without needing bolt slips.

(34) In an embodiment, the base connection **104** may include at least one shear tab **150** that is configured to connect the base plate **108** to the web **118** of the column **102**. The shear tab **150** is distinct from the structural fuse **112** and the flange plate **114**.

(35) As previously discussed, the structural fuse **112** is configured to yield when a sufficiently large load is applied to the structure **100**. FIG. 3A is a side elevational view of the structure **100** illustrated in FIGS. 1A-1D, according to an embodiment. FIG. 3B is a front elevational view of a yielded structural fuse **112**, according to an embodiment. A load L may be applied to the structure **100**. As illustrated in FIG. 5, the load L may cause the structure **100** to drift (i.e., sway) to the back. The load L may cause the back toe of the column **102** to bear on the base plate **108** which causes the front structural fuse **112** to yield. Likewise, when the structure **100** drifts to the front, the front toe of the column **102** bears on the base plate **108** which causes the back structural fuse **112** to yield.

(36) As illustrated in FIG. 3B, yielding the structural fuse **112** causes the yield regions **142** of the structural fuse **112** to yield. The yielded yield regions **142** are illustrated using non-intersecting diagonal lines. The yielded yield regions **142** may allow the structural fuse **112** to deform thereby absorbing and dissipating energy caused by the load L. However, the yielded yield regions **142** do not affect the attachment between the first connection region **136** and the column **102** and the second connection region **138** and the base plate **108** (e.g., the second connection region **138** and the flange plate **114** when the structural fuse **112** is connected to the base plate **108** using the flange plate **114**).

(37) After the structural fuse **112** yields, the structure **100** may be repaired. FIG. 4 is a flow chart of a method **400** for repairing the structure **100**, according to an embodiment. The method **400** includes block **405**, which recites “detaching the yielded structural fuse **112** from the column **102** and a base plate **108**.” In an embodiment, the yielded structural fuse **112** is bolted to at least one of the column **102** or the flange plate **114**. In such an embodiment, block **405** may include unbolting the yielded structural fuse **112** from the column **102** or the flange plate **114**. In an embodiment, the yielded structural fuse **112** is riveted to at least one of the column **102** or the flange plate **114**. In such an embodiment, block **405** may include grinding away, cutting, or otherwise removing the rivets such that the yielded structural fuse **112** may be detached from the column **102** or the flange plate **114**. In an embodiment, the yielded structural fuse **112** is welded to at least one of column **102**, the base plate **108**, or the flange plate **114**. In such an embodiment, detaching the yielded structural fuse **112** may include separating (e.g., cutting) the structural fuse **112** from the component(s) to which the yielded structural fuse **112** is attached or grinding away at least a

portion of the yielded structural fuse **112**.

(38) After block **405**, the method **400** may include block **410**, which recites “attaching a new structural fuse to the column **102** and the base plate **108**.” The new structural fuse may include any of the structural fuses disclosed herein. For example, the new structural fuse may be the same, substantially similar to, or different than the yielded structural fuse **112** before the yielded structural fuse **112** yielded. Attaching the new structural fuse to the column **102** and the base plate **108** may include at least one of bolting, riveting, welding, or otherwise attaching the new structural fuse to the column **102** and, either directly or indirectly via the flange plate **114**, to the base plate **108**. In an embodiment, the new structural fuse may be attached to the column **102** and the base plate **108** (e.g., indirectly via the flange plate **114**) using the same method as the yielded structural fuse **112**. In an embodiment, the new structural fuse may be attached to the column **102** and the base plate **108** using a method that is different than the yielded structural fuse **112**.

(39) The structural fuse illustrated in FIGS. **1A-3B** is merely one example of a structural fuse that may be included in the structures disclosed herein. Further examples of structural fuses that may be used in the structures disclosed herein are disclosed in U.S. Provisional Patent Application No. 63/174,706 filed on Apr. 14, 2021, U.S. Pat. No. 10,689,876 filed on Aug. 10, 2018, U.S. Pat. No. 10,584,477 filed on Apr. 25, 2019, U.S. Pat. No. 10,316,507 filed on Aug. 26, 2015, U.S. Pat. No. 10,760,261 filed on Dec. 8, 2016, and International Application No. WO 2021/030111 filed on Aug. 5, 2020, the disclosures of each of which are incorporated herein, in its entirety, by this reference.

(40) FIG. **5** is a side elevation view of a structure **500** with a load **L** applied thereto, according to an embodiment. The structure **500** includes a plurality of columns **502** and a plurality of beams **550**. The plurality of columns **502** and the plurality of beams **550** are connected together to form the structure **500**. The columns **502** may be secured to a foundation **506** using one or more base connections.

(41) As previously discussed, one or more portions of the structure **500** may be configured to yield to absorb and dissipate energy from the load **L** applied to the structure **500** to prevent catastrophic failure of the structure **500**. Examples of portions of the structure **500** that are configured to yield are indicated with circles on FIG. **5**. As shown, in some embodiments, one or more portions of the beams **550** adjacent to the columns **502** and/or the beam-to-column connections are configured to yield to absorb and dissipate energy. In some embodiments, the columns **502** proximate to the foundation **506** or the base connections (not shown) that connect the columns **502** to the foundation **506** are configured to yield to absorb and dissipate energy. At least one of the base connections that secure one or more of the columns **502** to the foundation **506** is the same or substantially similar to any of the base connections disclosed herein that include a structural fuse. As such, the structural fuse of the base connection yields instead of the column **502**. However, in some examples, at least one of the base connections that attach the columns **502** to the foundation **506** does not include a structural fuse. In such examples, at least one of the columns **502** yields, a portion of the base connections (e.g., the base plate or anchor rods) yields, or bolts slip in a slotted flange plate to absorb or dissipate energy from the load **L**.

(42) In an embodiment, at least a portion of the structures disclosed herein (e.g., the base connections disclosed herein) may be provided as a kit. The kit may include at least a portion of base connection. The base connection may include, for example, the base plate, one or more anchoring rods, and the structural fuse. The kit may also include one or more nuts for the anchor rods. The kit may further include at least one flange plate when the structural fuse is attached to the base plate using the flange plate. The flange plate may be provided as being attached (e.g., welded) to the base plate or separate from the base plate. The kit may also include the column to which the base connection is attached. The kit may be provided with one or more components thereof attached together and/or one or more components thereof not attached together. The components of the kit may be the same or substantially similar to any of the components (e.g., columns, base

plates, anchor rods, structural fuses, flange plate, etc.) disclosed herein.

(43) While various aspects and embodiments have been disclosed herein, other aspects and embodiments are contemplated. The various aspects and embodiments disclosed herein are for purposes of illustration and are not intended to be limiting.

(44) Terms of degree (e.g., “about,” “substantially,” “generally,” etc.) indicate structurally or functionally insignificant variations. In an example, when the term of degree is included with a term indicating quantity, the term of degree is interpreted to mean $\pm 10\%$, $\pm 5\%$, or $\pm 2\%$ of the term indicating quantity. In an example, when the term of degree is used to modify a shape, the term of degree indicates that the shape being modified by the term of degree has the appearance of the disclosed shape. For instance, the term of degree may be used to indicate that the shape may have rounded corners instead of sharp corners, curved edges instead of straight edges, one or more protrusions extending therefrom, is oblong, is the same as the disclosed shape, etc.

Claims

1. A base connection, comprising: a base plate including a top surface and a bottom surface opposite the top surface, the top surface configured to be adjacent to a terminal end of a column and the bottom surface configured to be adjacent to a foundation; one or more anchor rods attached to the base plate, the one or more anchor rods configured to secure the base plate to the foundation; at least one structural fuse configured to be attached to the column and attached to the base plate, the at least one structural fuse including a plate with at least one cutout formed therein, the at least one cutout configured to form one or more yield regions extending therefrom that are configured to preferentially yield relative to other regions of the plate; and at least one flange plate directly attached or integrally formed with the base plate, the at least one flange plate is distinct and separate from the at least one structural fuse, the at least one flange plate attachable to and detachable from the at least one structural fuse, wherein the at least one flange plate is oriented perpendicularly relative to the top surface of the base plate.
2. The base connection of claim 1, wherein the at least one structural fuse includes two structural fuses, each of the two structural fuses secured to one of two flanges of the column.
3. The base connection of claim 1, wherein the plate of the at least one structural fuse exhibits a thickness of about 0.5 cm to about 8 cm.
4. The base connection of claim 1, wherein each of the one or more yield regions extend from the at least one cutout in a first direction.
5. The base connection of claim 1, wherein the at least one cutout includes a plurality of cutouts separated by a spacing of about 5 cm to about 15 cm.
6. The base connection of claim 1, wherein the at least one cutout exhibits a maximum length and a width measured perpendicular to the maximum length that is less than the length, the width is about 2.5 cm to about 25 cm.
7. The base connection of claim 1, wherein the at least one structural fuse are bolted to the column.
8. The base connection of claim 1, wherein the at least one structural fuse is welded to the column.
9. The base connection of claim 1, wherein the at least one structural fuse directly contacts the column.
10. The base connection of claim 1, wherein the at least one flange plate at least partially defines at least one opening, the opening providing access to at least one region of the at least one structural fuse, wherein the at least one flange plate completely encloses the opening or exhibits a U-like shape.
11. The base connection of claim 10, wherein: the at least one structural fuse is configured to be bolted to the column; and the at least one opening provides access to portions of the at least one structural fuse that are bolted to the column.
12. The base connection of claim 1, wherein the base connection is a fully-restrained connection.

13. The base connection of claim 1, wherein the one or more anchor rods include a first end; and wherein a portion of the one or more anchor rods extending from the first end is disposed in and surrounded by the foundation.

14. The base connection of claim 1, wherein the at least one structural fuse includes at least one first connection region and at least one second connection region that is distinct and separate from the at least one first connection region, the at least one first connection region configured to be connected to the column and the at least one second connection region configured to be connected to the at least one flange plate.

15. The base connection of claim 14, wherein the at least one flange plate at least partially defines at least one opening, the opening providing access to the at least one first connection region, wherein the at least one flange plate completely encloses the opening or exhibits a U-like shape.

16. A method of repairing a yielded structural fuse, the method comprising: detaching the yielded structural fuse from a column and a base plate, the base connection including: the base plate including a top surface and a bottom surface opposite the top surface, the top surface adjacent to a terminal end of the column and the bottom surface adjacent to a foundation; one or more anchor rods attached to the base plate, the one or more anchor rods securing the base plate to the foundation; and at least one structural fuse including a plate with at least one cutout formed therein, the at least one cutout configured to form one or more yield regions extend therefrom that are configured to preferentially yield relative to other regions of the plate, the at least one structural fuse including the yielded structural fuse; and attaching a new structural fuse to the column and the base plate, the new structural fuse including a new plate with at least one new cutout formed therein, the at least one new cutout configured to form one or more new yield regions extending therefrom that are configured to preferentially yield relative to other regions of the plate wherein the base connection further includes at least one flange plate integrally formed with or configured to be attached to the base plate, the at least one flange plate is oriented perpendicularly relative to the top surface of the base plate, the at least one flange plate configured to be attached to the at least one structural fuse, wherein detaching the yielded structural fuse from the base connection include detaching the yielded structural fuse from the at least one flange plate and attaching the new structural fuse to the column and the base plate includes attaching the new structural fuse to the at least one flange plate.

17. The method of claim 16, wherein: detaching the yielded structural fuse from the base connection includes unbolting the yielded structural fuse from the column; and attaching the new structural fuse to the column and the base plate including bolting the new structural fuse to the column.
